

PEPETI BALAJI

Software Development Engineer in Test (SDET)

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SUMMARY

SDET with 2 years of experience building scalable UI, API, and performance automation for cloud-based product platforms. Skilled in Playwright/TypeScript, REST API and accessibility testing, and CI/CD pipeline ownership. Used AI-assisted engineering tools to raise automation throughput by 45% and to accelerate defect triage. Comfortable reading and modifying application code to isolate root causes, working directly with developers and PMs across the release cycle.

TECHNICAL SKILLS

Languages: Java 21, TypeScript, JavaScript, Python, SQL
UI & Accessibility Automation: Playwright, Selenium WebDriver, Axe DevTools (@axe-core/playwright), Test Data Builders
API & Performance Testing: REST API Testing, Postman, RestAssured, k6, Apache JMeter
CI/CD & DevOps: GitHub Actions, GitLab CI/CD, Jenkins, Docker, Docker Compose
Backend & Observability: Spring Boot 3, Apache Kafka, Redis, PostgreSQL, Prometheus, Grafana, Zipkin
Practices: Functional & Regression Testing, Defect Triage, Flaky-Test Resolution, Root-Cause Debugging
AI-Assisted Engineering: OpenAI Codex, GitHub Copilot, Cursor, Oracle Code Assist, Cline

PROFESSIONAL EXPERIENCE

QA Engineer — Test Automation & Backend Validation

Nov 2024 – Present

Oracle | Bengaluru, India

- Test framework ownership:** built and maintained modular Playwright + TypeScript automation suites, growing coverage to 500+ UI/UX and core learning workflows.
- AI-accelerated delivery:** adopted OpenAI Codex and AI-assisted coding tools into the test-authoring workflow, lifting sprint automation throughput by 45%.
- CI/CD reliability:** ran daily batch executions in CI pipelines with automated failure reporting, and drove root-cause analysis that eliminated recurring flaky UI tests.
- Cross-functional delivery:** partnered with PMs and developers each build cycle to clarify requirements, surface edge cases, and align on acceptance criteria.
- Quality gatekeeping:** performed end-to-end feature verification, filed developer-ready Jira bugs, and shipped small fixes directly to application code.
- Root-cause debugging:** traced system calls and database mutations through interactive local debugging sessions to resolve hidden logic defects.
- Accessibility compliance:** built automated WCAG scans with @axe-core/playwright, catching accessibility defects before release.
- Performance engineering:** designed k6 load and latency benchmarks for high-concurrency API endpoints to validate SLA compliance.

TECHNICAL PROJECTS

Event-Driven E-Commerce Microservices Platform ([repo](#))

Core Developer & SDET

Java 21, Spring Boot, Kafka, gRPC, Docker, Cloud, k6

- Built a production-scale distributed microservices backend secured end-to-end with OAuth2/OIDC token lifecycle management.
- Implemented real-time Kafka event streaming and Redis-backed state persistence for checkout and inventory allocation flows.
- Wrote API and system integration test suites covering gRPC contracts, RBAC rules, and failure-isolation policies; ran k6/JMeter load tests and set up Prometheus, Grafana, and Zipkin observability.

Library Management System ([repo](#))

Backend Developer & Test Engineer

Java, Spring Boot, JWT, PostgreSQL

- Built a modular microservices platform with decoupled routing, dynamic service registration, and lending-state management.
- Wrote unit and integration test suites with JUnit 5, Mockito, and MockMvc to catch interface mismatches early.
- Validated routing, catalog mutation, and stateless JWT authorization paths for zero-leak backend security.

EDUCATION & CERTIFICATIONS

B.Tech, Computer Science — Gayathri Vidya Parishad College of Engineering, Visakhapatnam (Mar 2021 – May 2024)

Diploma, Computer Engineering — AANM & VVRSR Gudlavalleru Polytechnic (Jul 2018 – May 2021)

Oracle Cloud Infrastructure 2024 Certified AI Foundations Associate — Oracle (2024)